Microsoft Azure for AI and Machine Learning

Module 1
Azure AI & ML environment setup

Module 2
Data preparation and model training in Azure

- Data preparation and model training in Azure — Video • 4 min
- Guide to creating ingestion pipelines — Reading • 5 min
- Practice activity: Creating an ingestion pipeline — Reading • 30 min
- Reflection: Creating an ingestion pipeline — Practice Assignment • Grade: 100%
- Walkthrough: Creating an ingestion pipeline (Optional) — Video • 6 min
- Knowledge check: Creating an ingestion pipeline — Practice Assignment • Grade: 80%
- Data preprocessing — Video • 5 min
- Explanation of preprocessing techniques — Reading • 10 min
- Practice activity: Implementing preprocessing techniques — Reading • 45 min
- Reflection: Implementing preprocessing techniques — Practice Assignment • Grade: 100%
- Walkthrough: Implementing preprocessing techniques (Optional) — Video • 7 min
- Model training — Video • 5 min
- How to train models using Azure Machine Learning — Video • 7 min
- Knowledge check: Model training — Practice Assignment • Grade: 100%
- Monitoring and logging training processes — Video • 4 min
- Explanation of monitoring and logging — Reading • 10 min
- Practice activity: Logging — Reading • 30 min
- Reflection: Logging — Practice Assignment • Grade: 100%
- Walkthrough: Implementing logging in ML systems (Optional) — Video • 5 min
- Summary: Data preparation and model training in Azure — Reading • 5 min
- Graded quiz: Data preparation and model training in Azure — Graded Assignment • 20 min

Module 3
Model deployment and management in Azure

Module 4
Troubleshooting Azure AI/ML workflows

Module 5
Toward systems integration

Summary: Data preparation and model training in Azure

Introduction

Machine learning models are only as good as the data they are trained on. Preparing high-quality data and effective training models are the backbone of every successful AI project. Azure provides an array of tools and services that simplify these processes, enabling data scientists to focus on building impactful solutions rather than grappling with infrastructure challenges.

Imagine being able to clean and transform raw data at scale, engineer new features seamlessly, and train models on distributed resources—all within an integrated ecosystem. With Azure's tools for data preparation and model training, this vision becomes a reality.

By the end of this reading, you'll be able to:

- Explain the importance of data preparation and model training in the machine learning lifecycle.
- Explore Azure's tools, such as Azure Data Factory, Azure Databricks, and Azure Machine Learning Studio, for streamlining data preparation and training.
- Learn best practices for ensuring data quality, feature engineering, and efficient model training.
- Discover how Azure's scalable resources and automated tools support hyperparameter tuning and reproducibility.
- Apply these insights to a real-world example of demand forecasting in retail.

Data preparation in Azure

Data preparation involves gathering, cleaning, and transforming raw data into a suitable format for training machine learning models. Azure offers several powerful tools for this stage:

- **Azure Data Factory:**
 - Create data pipelines for efficient data movement and transformation.
 - Automate data ingestion, preparation, and orchestration to ensure readiness for analysis.
- **Azure Databricks:**
 - Use this collaborative, Apache Spark-based analytics service for data wrangling, cleaning, and feature engineering at scale.
- **Azure Synapse Analytics:**
 - Perform large-scale data integration and preparation with data warehousing capabilities to streamline machine learning workflows.
- **Azure Machine Learning Studio:**
 - Leverage tools such as Data Wrangler for interactive, user-friendly data preparation with visualized transformations.

Best practices for data preparation

- **Data quality checks:** ensure data is accurate, complete, and consistent using Azure Data Factory or Databricks.
- **Feature engineering:** engineer new features to enhance model performance, such as encoding variables or aggregating values.
- **Automated data transformation:** minimize errors and ensure consistency across datasets through automated workflows.

Model training in Azure

Model training involves teaching an machine learning model to make predictions using input data. Azure's integrated ecosystem supports scalable and efficient model training:

- **Azure Machine Learning Studio:**
 - Manage the model training lifecycle using frameworks such as TensorFlow, PyTorch, and Scikit-Learn.
- **Azure Machine Learning Compute:**
 - Leverage managed clusters of virtual machines for distributed training of large-scale models.
- **HyperDrive:**
 - Automate hyperparameter optimization to find the best model configurations efficiently.
- **Integration with Azure Databricks:**
 - Use Databricks for distributed training on large datasets, enhancing scalability and speed.

Best practices for model training

- **Experiment racking:** record hyperparameters, metrics, and code versions to ensure reproducibility.
- **Scalable compute:** use Azure ML Compute to scale resources dynamically during training.
- **Early stopping:** halt training once validation performance stops improving to avoid overfitting.

Real-world example

Retail: Demand forecasting

A retail company aiming to forecast product demand can:

- Use **Azure Data Factory** to pull historical sales data from diverse sources.
- Process and clean the data using **Azure Databricks**.
- Train a predictive model in **Azure Machine Learning Studio**, optimizing it with **HyperDrive**.
- Deploy the model to inform inventory management decisions.

Conclusion

Data preparation and model training are critical steps in building effective machine learning solutions. Azure's ecosystem simplifies these tasks with tools such as Azure Data Factory for data transformation and Azure ML Compute for distributed training. By adopting best practices such as automated data transformation, experiment tracking, and hyperparameter tuning, data scientists can develop robust, scalable models that deliver meaningful results.

Mark as completed

Like Dislike Report an issue

Go to next item →

coach

Hi, Venkatesh!
How can I help?

Explain this topic in simple terms | Give me a summary | Give me practice questions | Give me real-life examples

Ask me anything

Coach is powered by AI, so check for mistakes and don't share sensitive info. Your data will be used in accordance with Coursera's Privacy Notice.